ORTA DOĞU TEKNİK ÜNİVERSİTESİ
MIDDLE EAST TECHNICAL UNIVERSITY

METU POWER LAB

EE561 Advanced Static Power Conversion

Multilevel Converters: Building the Levels — Quiz

Name: Student ID: Date:

Answer both questions in the space provided. Each question is worth 25 points (total 50). Both questions cover the basics of the talk: what a multilevel leg is, its advantages and disadvantages, and the topologies of the talk.

Question 1 · Levels

CONCEPT AND NUMBERS **25 pts**

A leg with N levels connects its output to one of N equally spaced DC voltages. The DC link is V_{dc} = **800 V**.

a. Complete the table. [15]

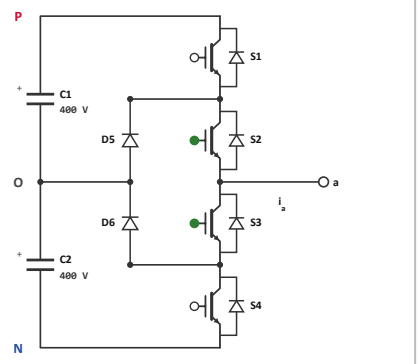
Number of levels N	**Voltage step = voltage blocked by each switch [V]**	**Number of switches in one leg**
2		
3		
5		

b. Give **two advantages** and **two disadvantages** of a multilevel leg compared with a two-level leg. [10]

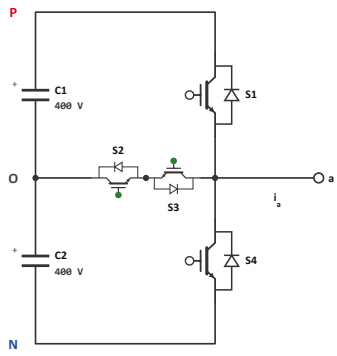
Question 2 · Reading the circuits

CIRCUITS **25 pts**

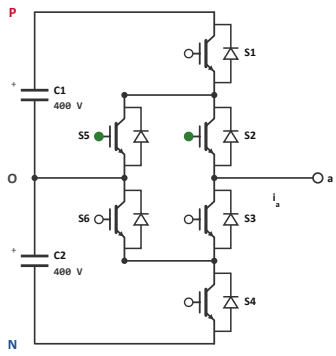
a. Each leg below is in its **0 V state** (gated switches green) and the current flows **out of the leg** ($i_a > 0$). **Draw the current path** on each figure, from the DC link to the output a, and write the name of the leg under it. [15]



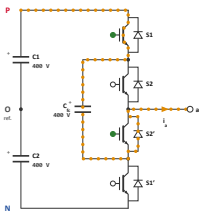
Leg A:

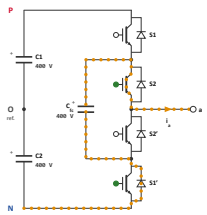


Leg B:



Leg C:





D (left) and **E** (right): gated switches green, current path dotted

b. Figures D and E show a flying-capacitor leg in two states. Both give 0 V at the output. What are such states called, and what is different for the capacitor C_{fc}? [10]